

EVORA, Portugal

WMO: 085570

Lat: 38.567N

Lon: 7.900W

Elev: 321

StdP: 97.53

Time Zone: 0.00 (EUW)

Period: 82-92

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	2.7	4.0	-4.3	2.7	6.8	-2.6	3.2	8.2	12.1	11.0	10.4	10.0	4.4	320	0.512

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.0	35.7	20.0	33.7	19.2	31.8	18.8	20.8	32.4	20.1	31.0	19.5	29.4	3.4	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.9	13.3	21.3	17.1	12.7	21.4	16.4	12.2	20.9	61.2	32.6	58.9	31.1	56.7	29.7	23.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.2	9.0	8.2	1.1	38.1	1.7	1.4	-0.2	39.1	-1.2	39.9	-2.2	40.8	-3.4	41.8		
			-0.7	22.0	1.8	0.7	-2.0	22.5	-3.0	22.9	-4.0	23.3	-5.3	23.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	16.0	9.2	10.3	12.5	13.3	16.4	19.9	23.3	23.4	22.5	17.4	13.1	10.5
	DBStd	5.90	2.16	2.69	2.82	3.09	3.48	3.48	3.74	3.12	3.47	3.19	2.78	2.22
	HDD10.0	105	40	26	7	5	1	0	0	0	0	0	7	20
	HDD18.3	1412	284	226	181	153	82	21	2	1	4	58	158	243
	CDD10.0	2298	14	34	86	104	197	298	413	414	375	229	99	35
	CDD18.3	563	0	0	1	2	21	68	157	157	129	28	1	0
	CDH23.3	5678	0	0	4	13	209	721	1803	1613	1153	161	1	0
	CDH26.7	2442	0	0	0	0	50	277	888	734	469	23	0	0
Wind	WSAvg	4.3	4.1	4.3	4.7	4.7	4.4	4.4	4.6	4.1	4.0	4.1	4.1	
Precipitation	PrecAvg	638	88	80	54	57	48	26	7	5	28	72	82	89
	PrecMax	1040	342	208	209	175	121	105	73	73	77	217	245	248
	PrecMin	367	1	4	0	4	0	0	0	0	1	2	4	4
	PrecStd	178	79	63	49	39	36	28	15	13	22	60	67	74
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.4	19.2	23.6	25.1	30.6	34.4	37.9	37.5	36.6	29.0	21.8	17.7
		MCWB	11.3	12.9	14.4	14.7	17.1	18.5	20.6	20.7	20.3	16.9	14.8	12.3
	2%	DB	14.2	17.0	21.5	23.0	27.9	32.1	35.9	35.2	33.8	26.9	19.7	15.8
		MCWB	11.0	12.0	13.6	14.1	15.9	18.8	20.2	19.9	18.8	17.0	14.7	13.1
	5%	DB	13.2	15.2	19.4	21.0	25.8	29.9	33.9	32.7	31.5	25.2	18.2	14.8
		MCWB	10.8	11.3	12.7	13.3	15.5	17.9	19.6	19.0	18.3	15.9	14.4	12.6
	10%	DB	12.5	14.0	17.8	19.2	23.2	27.6	31.8	30.7	29.3	23.1	16.8	13.7
		MCWB	10.3	11.0	12.1	12.7	14.7	17.3	19.2	18.4	17.9	15.5	13.9	11.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.2	14.5	15.5	16.2	17.8	20.5	22.1	21.5	20.7	18.7	18.2	15.6
		MCDB	14.0	17.2	20.4	21.7	27.4	30.8	33.2	33.9	31.0	25.7	18.8	16.0
	2%	WB	12.4	13.4	14.3	15.1	16.7	19.4	21.1	20.6	20.0	17.8	16.1	14.4
		MCDB	13.2	15.2	19.4	20.3	25.4	29.6	32.9	32.7	28.8	23.6	17.7	15.0
	5%	WB	11.5	12.6	13.4	14.2	16.0	18.6	20.3	19.9	19.3	17.1	15.3	13.2
		MCDB	12.5	14.2	18.3	18.8	23.9	28.2	31.8	30.7	27.6	21.8	17.0	14.1
	10%	WB	10.7	11.7	12.6	13.5	15.2	17.8	19.6	19.1	18.7	16.4	14.6	12.2
		MCDB	12.0	13.3	16.7	17.7	21.7	26.2	30.3	28.4	26.6	20.8	16.4	13.2
Mean Daily Temperature Range	5% DB	MDBR	5.9	6.1	7.9	8.0	9.5	11.2	13.0	12.6	10.6	7.9	5.8	5.2
		MCDBR	6.7	7.7	10.4	10.9	13.5	15.2	14.8	14.9	13.1	11.0	7.4	6.2
		MCWBR	4.4	4.3	4.9	4.9	5.0	5.3	4.5	4.8	4.4	4.0	3.6	4.0
	5% WB	MCDBR	5.2	6.3	9.1	9.3	11.9	13.6	14.3	13.8	11.1	8.7	5.4	4.2
		MCWBR	4.0	4.2	4.8	4.5	4.8	5.1	4.8	4.9	4.2	3.9	3.2	3.5
Clear-Sky Solar Irradiance	taub	0.309	0.324	0.354	0.363	0.371	0.374	0.355	0.371	0.367	0.345	0.321	0.309	
	taud	2.510	2.466	2.390	2.371	2.377	2.383	2.413	2.376	2.408	2.499	2.523	2.518	
	Ebn at Noon	873	903	905	914	909	902	917	895	879	868	851	845	
	Edh at Noon	77	92	110	119	120	120	115	117	107	88	76	72	
All-Sky Solar Radiation	RadAvg	2.31	3.26	4.37	5.50	6.65	7.53	7.82	6.93	5.34	3.67	2.56	2.06	
	RadStd	0.30	0.41	0.49	0.52	0.59	0.42	0.25	0.24	0.34	0.32	0.32	0.24	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (18 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page